Does Tightening Rent Regulation Reduce Housing Quality? Evidence from New York City

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Abstract

Modern rent regulations often seek to preserve landlords' maintenance incentives by allowing rent increases in response to investment. I study how dismantling these incentives affects housing quality and capital investment, exploiting New York's 2019 Housing Stability and Tenant Protection Act (HSTPA)–a landmark reform that sharply curtailed cost recovery for property improvements. Combining detailed building-level administrative data on code violations, tenant complaints, and building permit filings, and a difference-in-differences design, I find that tightening rent regulation leads to substantial deterioration in physical living conditions. Immediately hazardous code violations rise by 25% and tenant complaints increase by 8%. This deterioration is amplified among highly leveraged properties and falls disproportionately on low-income and minority neighborhoods. To trace the mechanisms behind this decline, I use a large language model to classify text descriptions of over 150,000 building alteration filings. I show that landlords sharply pull back on apartment renovations (-46.3%), while shifting toward regulatory arbitrage, increasing the number of apartment combinations that allowed owners to reset regulated rents. These findings suggest that cost-recovery channels are critical for sustaining housing quality under price controls, particularly for vulnerable tenants.

JEL Codes: R31, R38, L85

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1 Introduction

Housing affordability has become a central policy issue in high-demand housing markets (Galster and Lee, 2021; Baum-Snow and Duranton, 2025). Policymakers in many cities and states have responded by enacting or strengthening rent regulation (Ahern and Giacoletti, 2026).[1] A long-standing concern with these policies is that if rents cannot reflect investments in quality, landlords have less incentive to invest in and maintain their properties (Friedman and Stigler, 1946). The empirical literature on rent control has established quality as a core margin of landlord response (Kholodilin, 2024), but credibly identifying the causal effect of rent regulation on quality has been difficult.

This paper provides causal evidence on that question by studying the effects of New York's 2019 Housing Stability and Tenant Protection Act (HSTPA), a major rent stabilization reform, on housing quality and investment in New York City. Before the law, landlords could raise rents after vacancies, recover capital-improvement costs through permanent rent increases, and eventually remove high-rent units from stabilization. HSTPA eliminated vacancy decontrol, sharply curtailed vacancy increases, and made capital-improvement recovery smaller and temporary, aiming to limit rent increases for tenants in rent-stabilized apartments. The scale of this reform, along with the availability of detailed administrative data on New York City apartment buildings, makes this a well-suited setting for studying how strengthening rent regulation affects housing quality and investment.

To study the policy's effects, I assemble a rich building-level panel of New York City administrative records from 2016 through 2025. The data combine building characteristics and rent stabilization status with housing code violations, tenant complaints, building alterations, ownership, and mortgage records. These data let me study multiple channels through which rent stabilization may affect housing quality and investment. In particular, I examine the policy's effects on housing code violations, tenant complaints, and building alterations. Empirically, I compare the evolution of outcomes in rent-stabilized and non-stabilized buildings before and after HSTPA using a difference-in-differences design. The building-level panel lets me include building fixed effects, which absorb time-invariant building characteristics, along with flexible controls for neighborhood, building age, size, and type, and pre-policy housing quality. Identification therefore comes from within-period comparisons between stabilized and non-stabilized buildings, after controlling for trends across neighborhood and property-characteristic groups.

I find that HSTPA worsened measured housing quality. Immediately hazardous (Class C)

[1] Recent examples include Oregon's SB 608, enacted in 2019, which limited annual rent increases to 7 percent plus CPI, California's 2019 Tenant Protection Act, which capped annual increases at 5 percent plus local CPI or 10 percent, and St. Paul's 2021 rent stabilization ordinance (Ahern and Giacoletti, 2026).

violations, the most severe, increased by 0.9 violations per 100 units per six-month period, a 25% increase relative to the pre-HSTPA mean among stabilized buildings. Tenant complaints also increased by 8%. The estimated increases in Class C violations are concentrated in lower-income neighborhoods and neighborhoods with higher non-white population shares: 1.22 per 100 units in low-income tracts versus 0.10 in high-income tracts, and 1.39 in neighborhoods in the top tercile of non-white population share versus 0.28 in the bottom tercile.

To study effects on investment, I use a large language model to classify the free-text descriptions of 152,751 alteration projects filed with the Department of Buildings by the type of work each describes, such as renovating apartments, combining apartments, or replacing building-wide systems. I estimate that HSTPA led to a 46.3% decline in apartment renovations in stabilized buildings. I find no change in building-wide work that landlords could previously pass through to stabilized rents (-0.1%). Finally, I find a large increase in apartment combinations in stabilized buildings (159.4%).

I also examine whether highly leveraged landlords drive these effects on quality. By limiting landlords' ability to recover investment costs, HSTPA may have placed greater pressure on owners with larger existing debt obligations. I examine this possibility using pre-HSTPA mortgage exposure measured from New York City's Automated City Register Information System (ACRIS). I find that immediately hazardous violations increase significantly in every mortgage-exposure group, but that among buildings with qualifying mortgages, code violations rise more in those with larger mortgage exposures. This pattern suggests that while leverage may amplify deterioration in response to rent stabilization, it does not fully explain the aggregate effect.

This paper contributes to a broad literature studying the effects of rent regulation on landlords, tenants, and housing markets. Diamond et al. (2019) find that San Francisco's 1994 rent-control expansion reduced displacement among covered tenants but also reduced rental supply through conversions and redevelopment. Autor et al. (2014) show that ending rent control in Cambridge increased property values and investment, with spillovers to nearby units. In Germany, Baye and Dinger (2024) find that a rent brake reduced returns on regulated rental properties and shifted returns toward exempt properties. Ahern and Giacoletti (2026) show that St. Paul's 2021 rent-control ordinance reduced property values and generated uneven wealth effects across the income distribution of renters. Rent control also generates tenant-unit misallocation and price spillovers to the unregulated segment (Glaeser and Luttmer, 2003; Mense et al., 2023).

A smaller body of literature studies housing quality directly, though it often faces measurement and identification challenges. Gyourko and Linneman (1990) show that New York City's old rent controls were associated with lower rental housing quality, while Moon

and Stotsky (1993) use a structural model to show that greater rent suppression in New York City was associated with faster deterioration. Sims (2007) finds that decontrol in Massachusetts reduced maintenance problems, though primarily for minor deficiencies. More recently, Breidenbach et al. (2022) find that Germany's rent brake lowered the probability that treated rental listings were advertised in good condition. The closest previous work to my paper is Seltzer (2024), who studies how New York's 2011 Rent Act, which reduced cost recovery for apartment improvements in larger rent-stabilized buildings, affected housing code violations. Using housing-code violations and a matched difference-in-differences design, he finds that violations increased after the reform, especially among buildings whose owners were financially constrained. My paper builds on that work by studying a broader restructuring of the rent-stabilization system, combining the data with tenant-complaint data, and measuring investment directly by classifying the text of building-alteration filings with a large language model.

This paper also contributes to a literature on how the costs and benefits of New York City's rent stabilization system are distributed among tenants. Chen et al. (2023) document substantial rent discounts but find that the benefits are not progressively distributed and were historically larger for white tenants. Using survey measures of physical conditions together with code violations and 311 complaints, Hu et al. (2025) document racial disparities in rental housing quality and in quality per dollar of rent. By contrast, I study how tightening rent-stabilization changed landlords' maintenance and investment behavior, and which neighborhoods bore the resulting quality costs.

Policy groups and other stakeholders have raised concerns about the effects of HSTPA on housing quality, but prior evidence on HSTPA itself is mainly descriptive. NYU Furman Center (2021) examines short-run trends in complaints, violations, and alteration filings after HSTPA and finds little immediate change in complaints or violations, though alteration filings declined after an initial spike. More recently, NYU Furman Center (2026) documents deteriorating operating margins and larger post-2021 increases in code violations among legacy buildings with at least 90 percent rent-stabilized units. A survey of owners and managers commissioned by industry groups reports that owners viewed HSTPA as making apartment renovations harder to finance and documents sharp declines in applications for rent increases tied to building-wide and apartment-level improvements (HR&A Advisors, 2024). This paper provides causal evidence on this pertinent policy question.

The rest of the paper proceeds as follows. Section 2 describes HSTPA and the investment incentives it changed. Section 3 describes the building-level panel. Section 4 presents the empirical strategy. Section 5 reports the main results, heterogeneity by neighborhood and mortgage exposure, and robustness checks. Section 6 concludes.

2 Policy Background

2.1 Rent Stabilization in New York City

New York City's rent stabilization system covers approximately one million apartments, primarily in buildings with six or more units built before 1974.[2] The pre-HSTPA system had important features of what Arnott (1995) calls "second-generation" rent control, limiting rent increases while still allowing landlords to raise rents and recoup costs through several channels. The Rent Guidelines Board allowed annual rent increases, typically 1–3%. Landlords could also generally raise rents when a tenant left the unit. The system also allowed for specific rent increases to recover the cost of maintenance and investment. Landlords could raise rents to cover approved building-wide improvements, such as boilers, roofs, elevators, and windows, through what are called Major Capital Improvement (MCI) rent increases. Landlords could also recover the cost of unit-level renovations through Individual Apartment Improvement (IAI) increases, which were permanent and uncapped, with the amount varying by building size. Finally, the old system allowed units to leave rent stabilization when legal rents, and in some cases tenant incomes, exceeded certain thresholds. Together, these features at least partially preserved landlords' incentives to invest in and maintain stabilized buildings, both to increase rents and to eventually exit stabilization.

2.2 The Housing Stability and Tenant Protection Act (2019)

On June 14, 2019, New York enacted the Housing Stability and Tenant Protection Act (HSTPA). The law was designed to protect tenants and maintain the rent-stabilized housing stock by limiting large rent increases after turnover or improvements and reducing the ability of stabilized units to exit rent stabilization.[3]

HSTPA first weakened the link between tenant turnover and future rent growth. Before the law, a vacancy could raise the legal rent by up to 20% for a two-year lease, with additional longevity increases in some cases. HSTPA eliminated these statutory vacancy and longevity increases and barred the Rent Guidelines Board from setting separate vacancy adjustments. It also repealed high-rent and high-income deregulation. Before HSTPA, a unit could leave stabilization if its legal rent reached the deregulation threshold, which was $2,774 in New York City in 2019, either when the unit became vacant or, for occupied units, when tenant income exceeded $200,000 for two consecutive years. After HSTPA, rent increases no longer

[2]Exact counts vary depending on data source and year. The NYC Rent Guidelines Board estimated 966,000 stabilized units in 2018; see its rent-stabilized building lists.

[3]For an overview of the purpose of HSTPA and early debates around the program, see NYU Furman Center (2021), *Housing Stability and Tenant Protection Act: An Initial Analysis of Short-Term Trends*.

created the same path from investment to deregulation.[4]

HSTPA also sharply reduced landlords' ability to recover capital improvement costs by shrinking the rent increases owners could charge and, for both improvement channels, making those increases temporary rather than permanent. For building-wide Major Capital Improvements, the law capped the annual increase at 2% of existing rent and required owners to remove it after 30 years. For Individual Apartment Improvements, as enacted in 2019, it capped eligible costs at $15,000 over a 15-year period, limited owners to no more than three increases in that window, and cut the monthly rent-increase formula to 1/168 or 1/180 of eligible costs.[5]

Taken together, these legal changes moved the rent stabilization system away from the more flexible, second-generation regime that preceded HSTPA, reducing landlords' ability to raise rents in response to investments, and the option value of improving units toward deregulation. Using building-level data on housing code violations, tenant complaints, and building alterations, I estimate how these changes affected housing quality and investment in rent-stabilized buildings.

3 Data

3.1 Data Sources and Variables

I assemble administrative data from multiple New York City and State agencies to measure housing quality, investment, and rent stabilization status at the building level. I link datasets using the Borough-Block-Lot (BBL) identifier, a 10-digit code that uniquely identifies every tax lot in New York City.[6]

Building characteristics. The Primary Land Use Tax Lot Output (PLUTO) database, published by the Department of City Planning and available through NYC Open Data, provides the universe of tax lots and their characteristics.[7] Residential-unit counts, building class, and census tract come from PLUTO 18v2.1, which was released before HSTPA.

[4]For an overview of the changes, see NYS Homes and Community Renewal, *Strengthening New York State Rent Regulations: The Housing Stability and Tenant Protection Act of 2019*. For historical deregulation rent and income thresholds, see HCR Fact Sheet #36.

[5]For MCI rules, see HCR Fact Sheet #24. The Department of Housing and Community Renewal overview cited above summarizes the 2019 IAI changes. Later legislation modified the IAI rules effective October 17, 2024, raising the cap to $30,000 generally and up to $50,000 in specified cases, and making new IAI increases permanent; see HCR's summary of FY24 housing-law changes. These 2024 changes, which weaken restrictions on IAI, should reduce estimates of HSTPA's effects after 2024.

[6]BBL identifiers are unique to a tax lot, which can contain more than one physical building. I will use tax lot and building interchangeably throughout the paper.

[7]NYC Open Data is the city's public data portal, available at `https://opendata.cityofnewyork.us/`.

Rent stabilization status. Stabilization status comes from lists of rent-stabilized buildings compiled by the New York State Division of Housing and Community Renewal (DHCR), which administers rent stabilization. A building is on a year's list if its owner registered at least one stabilized unit with DHCR that year. The NYC Rent Guidelines Board publishes the lists as PDFs of tax lots by borough. I use two lists of rent stabilization status. For 2017, I use the digitized list archived by Maxwell Austensen, and for 2024 I digitize the PDFs myself.[8] I match both lists to tax lots by BBL. In my main specifications, I define treated buildings as buildings listed in 2017, two years before HSTPA passed, to avoid concerns that landlords pushed to evade stabilization in anticipation of the law. I drop buildings that are not listed as stabilized in 2017 but are listed in 2024 from the sample because I cannot establish their stabilization status at the time of the law.[9]

Housing quality outcomes. To measure tenant-facing housing quality, I use data on code violations and complaints, both from the New York City Department of Housing Preservation and Development (HPD) through NYC Open Data.

Code violations come from HPD's Housing Maintenance Code Violations dataset. HPD assigns each violation one of three hazard classes. Class A (non-hazardous) covers violations such as signage and recordkeeping problems. Class B (hazardous) covers conditions affecting health, safety, or habitability. Class C (immediately hazardous) violations are the most severe, such as heat and hot-water failures, lead-based paint hazards, mold, and pests. My analysis focuses on Class C violations because they capture serious quality issues that affect most tenants' well-being, but I also report results on Class B and Class A violations.

Tenant complaints come from HPD's Housing Maintenance Code Complaints and Problems dataset. Tenants file complaints through 311, online, or HPD Code Enforcement Borough Offices, describing conditions such as lack of heat, water leaks, or pest infestations. Complaints record what tenants report, whether or not HPD later verifies a code violation.

Investment outcomes. I measure investment with building alteration jobs filed with the Department of Buildings (DOB). I combine records from the department's legacy Building Information System (BIS) and its newer platform, DOB NOW.[10] I begin with the universe of alteration jobs (BIS types A2 and A3 and their DOB NOW equivalents) with no recorded change in the number of dwelling units.

A raw count of these filings, however, may be a poor measure of the investment response

[8]Lists at `https://rentguidelinesboard.cityofnewyork.us/resources/rent-stabilized-building-lists/`. The 2017 list is archived at `https://github.com/austensen/dhcr-rgb-buildings`.

[9]About half of these buildings had stabilized units on a pre-HSTPA tax bill, so they are not valid controls, and the rest have no pre-HSTPA record, so they are not clearly treated.

[10]DOB moved filing from BIS to DOB NOW during the sample period. The two systems record filings differently, and DOB NOW records later amendments as separate filings, so I reduce both to one record per job.

to HSTPA. Alteration filings include sidewalk sheds and scaffolds, inspections, legalizations of past work, and changes to commercial space, alongside apartment renovations and building-wide work. I therefore classify each filing by the work described in its free-text job description using a large language model (Claude Haiku 5.5). The model then assigns the filing to one of ten categories (apartment renovation, interior work with no stated location, building systems, envelope, structural, common areas, site work, commercial space, not investment, or unclear), records what kind of work the filing names, and flags filings that ride on another application ("filed in conjunction with") or respond to a violation, order, or facade-inspection cycle. Appendix B explains the classification rules in more detail.

From these classifications, I build four outcomes. *Apartment renovation* is work in or for identified apartments. *Apartment combinations* are renovations that merge apartments or rooms across apartments. *MCI-type work* is building-wide work (systems, envelope, structural, common areas, or site work) that names an item on DHCR's Useful Life Schedule, the list of improvements eligible for Major Capital Improvement rent increases.[11] Work on *commercial space* serves as a placebo, since HSTPA did not change the returns to commercial work.

In addition to these LLM-based investment measures, I collect data on Major Capital Improvement (MCI) and Individual Apartment Improvement (IAI) applications for descriptive analyses. I collect data on MCI applications filed in New York City through November 2020 from the NYCDB project.[12] For later years, I also collect statewide counts of MCI applications from the annual reports of DHCR's Office of Rent Administration.[13] I collect annual counts of IAIs registered in New York City from the Rent Guidelines Board's 2023 explanatory statement.[14]

Building-level mortgage exposure. For a supplementary heterogeneity analysis, I measure each building's mortgage exposure as the building's most recent pre-HSTPA mortgage origination amount divided by its pre-HSTPA market value. This serves as a proxy for each building's loan-to-value ratio (LTV). I pull mortgage origination data from New York City's Automated City Register Information System (ACRIS), a public database of property transactions and mortgage documents.[15] For each building, I divide the amount of its most

[11] DHCR Fact Sheet #33, "Useful Life Schedule for Major Capital Improvements," September 2019 version, `https://hcr.ny.gov/system/files/documents/2020/11/fact-sheet-33-09-2019.pdf`.

[12] The MCI application records were originally obtained from DHCR through a FOIL request by Winnie Shen and are available at `https://github.com/nycdb/nycdb`.

[13] Reports for 2020 through 2025, from `https://hcr.ny.gov/office-rent-administration-transparency-initiative`.

[14] Rent Guidelines Board, "Explanatory Statement and Findings of the Rent Guidelines Board in Relation to 2023–24 Lease Increase Allowances for Apartments and Lofts," Order #55, June 2023, p. 37, `https://rentguidelinesboard.cityofnewyork.us/wp-content/uploads/2023/06/2023-Apartment-Loft-Explanatory-Statement-55.pdf`.

[15] ACRIS data are available through NYC Open Data.

recent mortgage recorded between January 2010 and June 13, 2019, by its estimated market value on the Department of Finance FY2020 final property-tax roll, which was published on May 28, 2019, before HSTPA's passage. I winsorize this ratio at the 99th percentile and divide buildings with valid, positive ratios into three mortgage-exposure terciles. I keep buildings without an ACRIS mortgage during the 2010–2019 window as a separate group. ACRIS reports origination amounts rather than outstanding balances, so this measure ranks buildings by pre-HSTPA borrowing rather than measuring current leverage. I exclude Staten Island from these analyses because ACRIS mortgage data does not include its property records.

Building Ownership. Landlords may own both stabilized and non-stabilized buildings, and one possibility is that landlords may shift behavior across their portfolio in response to changes in rent stabilization. To identify buildings that share an owner, I link buildings to individual owners using HPD Multiple Dwelling Registrations data.[16] Many multifamily buildings are held through single-purpose LLCs, so the registered owner's name identifies the entity, but not the person behind it. However, HPD requires owners of buildings with three or more residential units to register each year and name the individual owner, or, for a corporate owner, its head officer, so it is possible to identify the building's true owner. I use that person for each building: the individual owner when listed and, otherwise, the joint owner, head officer, or officer. After standardizing names, I link buildings whose registrations name the same person. This methodology builds on the one used by JustFix's Who Owns What project (JustFix, 2026). My landlord matching procedure identifies the landlords of 40,046 buildings (97% of the sample)[17] and 24,668 distinct owners. 1,335 owners (5.4%) hold both stabilized and non-stabilized buildings in their portfolios. These registrations reflect current filings as of February 2026, so the linkage measures portfolios as of that date rather than at HSTPA's passage. Appendix C provides more details on the portfolio classification procedures.

Neighborhood characteristics. To explore heterogeneity by neighborhood characteristics, I collect tract-level median household income and non-white population share from the 2015–2019 American Community Survey 5-year estimates.

3.2 Sample and Summary Statistics

I restrict my analysis to multifamily buildings where rent stabilization is concentrated: buildings with six or more residential units that were built in 1990 or earlier. I drop lots

[16] I pull HPD Registration Contacts and Multiple Dwelling Registrations from NYC Open Data.

[17] The remaining 1,210 buildings (2.9%) have no usable owner name on their registration. They stay in the main analysis but are not used in owner-portfolio classifications.

with no valid construction year in PLUTO. I then exclude New York City Housing Authority public housing and tax lots linked to affordable housing subsidies before HSTPA. In particular, I exclude buildings linked to the Low Income Housing Tax Credit (LIHTC), Mitchell-Lama subsidies, and HUD project-based assistance. I collect these data from the New York City Housing Authority, HPD's record of affordable-housing production agreements, and the NYU Furman Center's Subsidized Housing Database. Next, as mentioned above, I drop buildings on the 2024 DHCR stabilized unit list but not the 2017 list, and I also exclude condominium lots (PLUTO building class R) and co-op buildings (PLUTO classes C6, C8, CC, D0, D4, DC, H7), where maintenance decisions rest with individual unit owners or a board of owner-occupants rather than a landlord. Table 1 summarizes these sample restrictions. The resulting sample contains 41,256 buildings, 31,066 stabilized and 10,190 control.

Table 2 presents summary statistics separately for rent-stabilized and non-stabilized buildings. Stabilized buildings contain an average of 27 units, compared with 17 for non-stabilized buildings. The average age of buildings in the two groups is similar, with construction years of 1921 and 1920 in the stabilized and non-stabilized sample, respectively. However, stabilized buildings have substantially higher baseline code violation rates (0.16 Class C violations per building per month vs. 0.06) and complaint rates (0.56 vs. 0.17 per building per month). Figure 1 shows the geographic distribution: rent-stabilized buildings are concentrated in Manhattan, the Bronx, and parts of Brooklyn and Queens, with substantial overlap between stabilized and non-stabilized buildings within neighborhoods.

4 Empirical Strategy

To estimate the effect of HSTPA, I use a difference-in-differences design that compares changes in outcomes for rent-stabilized buildings with changes for non-stabilized buildings before and after the law was passed in June 2019. The main identification challenge is that stabilized buildings and non-stabilized buildings differed substantially even before HSTPA. For example, stabilized buildings already had nearly three times as many Class C violations per building per month (0.16 vs. 0.06) and more than three times as many monthly complaints (0.56 vs. 0.17). These baseline differences do not necessarily invalidate the difference-in-differences design, but they raise the concern that stabilized and unstabilized buildings would have followed different trends even without HSTPA.

My empirical strategy therefore controls for a variety of factors to address these differences, leveraging the rich administrative New York City data. Building fixed effects absorb all time-invariant differences across buildings. The remaining controls are a set of group $\times$ period fixed effects, which absorb the average shock the whole group experienced in the period,

for five groups: tract, building size bin, construction decade, building type, and baseline quality. Tract × period fixed effects absorb shocks common to buildings in the same tract and period, such as changes in local housing demand, gentrification, enforcement intensity, and neighborhood-level pandemic disruptions. Size-bin × period, construction-decade × period, and building-type × period fixed effects allow outcomes to evolve differently across buildings of different sizes, vintages, and types.[18] Finally, I control for baseline building quality, since buildings in poor condition before the law may have followed different trajectories over time, regardless of their stabilization status. I compute each building's average number of Class C violations per six-month period over the three years before HSTPA and sort buildings into three groups: those with no Class C violations, those at or below the median count among buildings with any, and those above the median.[19] Baseline-quality-group × period fixed effects then allow outcomes to evolve differently across buildings that started in different condition. Identification comes from differences between stabilized and non-stabilized buildings after absorbing these group-specific time paths.

I aggregate observations into six-month periods because monthly violations, complaints, and alteration filings are sparse at the building level, and longer bins reduce noise and smooth seasonal patterns. HSTPA passed on June 14, 2019, so I use January–June 2019 as the final reference period and July–December 2019 as the first post-period. For code violations, complaints, and alteration filings, I normalize counts by building size and report them per 100 residential units. I weight each regression by the number of residential units, giving each unit equal weight rather than each building. Appendix Table A1 reports raw-count specifications that give each building equal weight, which produce similar conclusions.

For the extensive margin, I estimate the same specification with an indicator for any Class C violation in the period as the outcome, so coefficients are percentage points.

Formally, I estimate:

$$Y_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \cdot \text{Stabilized}_i \times \mathbf{1}[t = k] + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{quality}}_{q(i),t} + \varepsilon_{it}. \quad (1)$$

Here, Y_{it} is the outcome for building i in 6-month period t and α_i denotes building fixed effects. Stabilized_i equals one for rent-stabilized buildings. The γ terms are the additive group-by-period fixed effects indexed by tract c, size bin s, construction decade d, building type b, and baseline quality group q, defined from pre-HSTPA Class C violations and the same for every outcome. The omitted event-time period, $k = -1$, is January–June 2019, the

[18] I use size bins of 6–9, 10–19, 20–49, 50–99, and 100+ units. Construction-decade bins are pre-1920 and each decade from the 1920s through the 1980s, the last including buildings completed in 1990. The building types are walk-up apartments, elevator apartments, and mixed-use buildings.

[19] More than half of buildings have no Class C violations in the pre-period, so the first group is the largest.

final six-month bin assigned to the pre-period. Each coefficient β_k measures the differential change in the outcome for stabilized relative to non-stabilized buildings in period k, relative to the corresponding difference in the omitted period. I cluster standard errors at the BBL level to allow arbitrary serial correlation within buildings.

For average post-HSTPA effects, I also estimate the pooled difference-in-differences specification:

$$Y_{it} = \alpha_i + \beta \cdot \text{Stabilized}_i \times \text{Post}_t + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{quality}}_{q(i),t} + \varepsilon_{it}. \quad (2)$$

where $\text{Post}_t = \mathbf{1}[t \geq 0]$ indicates periods after HSTPA. The coefficient β measures the differential post-HSTPA change for stabilized relative to non-stabilized buildings, conditional on the same fixed effects as the event-study specification.

I also ask whether the effects of HSTPA differ by borough, by terciles of tract-level median income and non-white population share, and by building-level mortgage exposure. For each dimension, I estimate one pooled specification that interacts $\text{Stabilized}_i \times \text{Post}_t$ with the group indicators, so that each coefficient compares the post-HSTPA change in stabilized buildings with the change in non-stabilized buildings in the same group. For borough and neighborhood tercile regressions, the tract $\times$ period fixed effects already absorb each group's common shocks. In the mortgage exposure regressions, I add mortgage-exposure-group $\times$ period fixed effects, since mortgage exposure varies at the building level.

The key identifying assumption is conditional parallel trends: after conditioning on the fixed effects described above, stabilized and non-stabilized buildings would have experienced similar changes absent HSTPA. I assess the plausibility of this assumption using the pre-HSTPA event-study coefficients. Estimates close to zero indicate that the two groups were not already diverging before the reform, although they cannot establish what would have happened afterward without HSTPA.

The difference-in-differences design also assumes limited anticipation. HSTPA was debated before June 2019, so landlords may have adjusted their behavior before its passage. I argue that anticipation is unlikely to drive the results. The absence of sustained divergence in the pre-HSTPA event-study coefficients argues against large anticipatory responses by landlords. I do see a spike in MCI applications just before the law passed (Figure 5), suggesting landlords may have rushed to file capital improvements before the new caps. This may have left stabilized buildings in better condition, which, if anything, would bias the results against finding post-HSTPA deterioration. As discussed in Section 3, I define stabilized buildings using the 2017 DHCR registry, two years before the law's passage, to limit concerns that landlords may have tried to exit rent stabilization in anticipation of HSTPA.

A final concern relates to potential spillovers across buildings owned by the same landlord. For example, if landlords who own both stabilized and non-stabilized buildings reallocated resources toward their non-stabilized buildings after HSTPA, the control group would improve and the estimates would overstate deterioration in treated buildings. Section 5 addresses this concern by re-estimating the main specification among buildings linked to owners whose observed portfolios contain only stabilized or only non-stabilized buildings. This restriction removes detected common ownership across the two groups.

5 Results

5.1 Housing quality

I find that HSTPA worsened housing conditions in stabilized buildings. Figure 2 plots the event-study coefficients for Class C (immediately hazardous) violations, and Table 3 reports the pooled estimates. Class C violations rose by 0.9 per 100 units per six-month period, a 25% increase relative to the pre-HSTPA mean among stabilized buildings. Part of this increase is on the extensive margin. The share of units in buildings that record at least one Class C violation in a six-month period rises by 5.88 percentage points, from a pre-HSTPA baseline of 32.4% (Appendix Figure A1). More buildings record a serious violation after HSTPA, not only buildings that already had them. These effects are concentrated among the most severe code violations: the Class B (hazardous) estimate is 0.33 per 100 units, only 3% of its baseline and statistically imprecise. Appendix Figure A3 plots the event studies for Class B and Class A violations.

Figure 3 presents the event study for tenant complaints. One concern with the code violation estimates could be that these increases may reflect increased code inspection activity rather than actual changes in housing quality. However, I find that HSTPA also led to increases in tenant complaints. Table 3 shows that HPD complaints increased by 1.1 per 100 units per six-month period (8% of baseline).

5.2 Investment

I find evidence that landlords responded to HSTPA by reducing investment in their properties. Figure 4 and Table 4 report estimates for the four classified outcomes defined in Section 3, counting each project once at its initial filing. At first glance, I find little evidence that the overall count of alteration filings changes. However, the picture changes once I sort filings by the work they describe. I find that HSTPA led to a 46.3% decline in apartment renovations in stabilized buildings relative to the pre-HSTPA baseline.

By contrast, I find little evidence of a decline in what I define as MCI-type work, building-wide work on items eligible for Major Capital Improvements rent increases. Work on commercial space, which HSTPA did not touch, changes by -8.2%.

One other finding from the alteration filings is a dramatic increase in filings to combine apartments in stabilized buildings. Before November 2023, an owner who combined two vacant stabilized apartments could set a new first rent for the combined unit, above the sum of the two previous rents. Combinations rise by 159.4% from a pre-HSTPA baseline of 0.0055 filings per 100 units per period. This increase is concentrated between July 2019 and December 2023 (Figure 4), consistent with the rule change that ended first rents for combined apartments.[20]

I also inspect the raw counts of MCI and IAI applications over time. Figure 5 presents the time series. MCI applications filed with DHCR fell from 101 per month before HSTPA to 21 per month after, a 79% decline, and registered IAIs in New York City fell from 14,356 in 2018 to 3,342 in 2021.

5.3 Robustness

Reporting and enforcement. A major concern could be that rather than picking up changes in quality, my estimates are picking up changes in either tenant reporting behavior or in proactive code enforcement that differentially affected stabilized buildings. Three patterns argue against this reading.

First, complaints rose, not just actual violations. Complaints come from tenants and do not require an inspection, so the increase cannot simply reflect more frequent inspections alone.

Second, if HSTPA made tenants in rent-stabilized apartments more willing to report problems, the additional reporting should fall on conditions that tenants had previously tolerated. Class C violations, the immediately hazardous class, are the most likely to be reported in any case. Marginal reporting should instead show up most in Class B violations, which are hazardous but not immediately threatening. However, I find small and imprecise effects on these violations.

Third, the one enforcement change that occurred around the time of HSTPA does not explain the result. In January 2019, the city launched LeadFreeNYC, which added proactive audits of landlords' lead records and increased targeted inspections (City of New York,

[20]Before November 2023, an owner who combined two vacant stabilized apartments could set a new first rent for the combined unit. Amendments to the Rent Stabilization Code effective November 2023 set the initial rent of a combined apartment at the sum of the previous legal rents (New York State Division of Housing and Community Renewal, 2023).

2020). In December 2021, the city halved the concentration threshold for paint to count as lead-based (New York City Department of Housing Preservation and Development, 2021). Lead violations are Class C violations, so one concern is that this enforcement change drives my Class C results. To address this concern, I estimate the main specification separately for lead-based violations and all other Class C violations. Table A5 shows these effects. I find that non-lead Class C violations drive the results.

Fixed effects specifications. One potential concern would be that these results are generated by the particular set of fixed effects used in the main specification. Table A2 adds the five sets of group-by-period fixed effects one at a time. The estimate is positive and significant in every column, and the pre-period fit improves as I add controls.

Functional form. My main specifications estimate the effect of HSTPA on code violations per 100 units. As an alternative specification, I also estimate the proportional effect on code violations using a Poisson pseudo-maximum likelihood (PPML) model, which estimates the percentage change in violations directly. Figure 6 shows the event study from the PPML model. The pooled estimate implies that HSTPA led to a 10.1% increase in Class C violations in stabilized units.

This estimated effect is positive and statistically significant, but smaller than that implied in my main specification. Two factors could explain this discrepancy. First, with the rich set of fixed effects, PPML estimates effects only for buildings and group-by-period cells with at least one Class C violation. Thus, the sample size used to estimate the effect is notably smaller. Second, PPML weights each building by its expected count, so buildings with many pre-HSTPA violations dominate the proportional estimate, while the levels estimate counts each additional violation the same. Panel B of Table A3 estimates the effect within the baseline-quality groups. Among buildings with positive baseline Class C violations, PPML implies a 30.1% increase for buildings below the median and a 8.2% increase for buildings above the median. The corresponding level effects point the same way, at 1.40 and 0.19 additional violations per 100 units, and buildings with no Class C violations during the baseline period experience an increase of 1.15. The deterioration is largest, in both levels and proportions, among buildings that entered HSTPA with few or no serious violations. The PPML estimator, which weights the worst buildings more heavily, therefore provides a smaller estimate.

Portfolio spillovers. A landlord who owns both stabilized and non-stabilized buildings could change their maintenance of stabilized properties in response to HSTPA. For example, if they shifted investment toward their unstabilized properties, the control group would improve, and the estimates would overstate deterioration in the stabilized buildings. To test this, I link buildings to owners using HPD registration data, as described in Section 3, and re-estimate

the main specification after excluding all buildings owned by these mixed-portfolio owners and buildings that have no matched owner. The results are shown in Appendix C. The estimate rises to 1.40 Class C violations per 100 units, compared to 0.9 in the full sample. Thus, reallocation within mixed portfolios is not inflating the main estimate. This check removes detected common ownership. Because the registration snapshot is not fixed before HSTPA, it cannot establish the absence of historical portfolio spillovers.

COVID-19. The post-period includes the COVID-19 pandemic, which likely affected building inspections, tenant complaint behavior, and construction activity. Two considerations address any potential bias from COVID-related disruptions. First, the tract × period fixed effects absorb citywide and neighborhood-level disruptions common to stabilized and non-stabilized buildings. Second, as a robustness check, I exclude the COVID period entirely (January 2020 through December 2021). Appendix Table A4 reports the resulting estimates for each outcome, which show that the effects are not driven by the COVID-19 period.

5.4 Heterogeneity

The aggregate effects may mask important differences in where the deterioration occurred and which buildings were most affected. I find that quality deterioration was concentrated in poorer neighborhoods and those with more non-white residents. Table 5 presents pooled difference-in-differences estimates for Class C violations separately by borough and by tercile of tract income and non-white population share. The Bronx has the largest estimate, 2.19 violations per 100 units, against 0.60 to 0.87 in the other three boroughs. The income gradient is sharp. The estimate falls from 1.22 violations per 100 units in the lowest-income tercile of tracts to 0.10 in the highest, and the two differ ($p = 0.022$). By non-white share, it rises from 0.28 in the lowest tercile to 1.39 in the highest ($p = 0.037$ for the difference).

By limiting future rent growth and landlords' ability to recover investment costs, HSTPA may have placed greater financial pressure on buildings with larger debt obligations. I therefore examine whether the effects differ with buildings' pre-HSTPA mortgage exposure. Table 6 reports group-specific estimates. Class C violations rise by 0.61 violations per 100 units among buildings without a qualifying mortgage record, 0.35 in the low mortgage-exposure tercile, 1.32 in the middle tercile, and 1.54 in the high tercile. The low-exposure estimate is not statistically significant, but the other three are, and the high-exposure estimate is significantly larger than the low-exposure estimate. This pattern is consistent with mortgage exposure amplifying the deterioration in housing quality. At the same time, the aggregate effect is not a story about leveraged landlords alone.

6 Conclusion

This paper studies how New York's 2019 Housing Stability and Tenant Protection Act, which strengthened rent stabilization and reduced landlords' ability to raise rents in response to investment, affected housing quality and investment in New York City's rent-stabilized buildings. Using a difference-in-differences design with a rich set of controls, I find that the policy substantially increased hazardous code violations and tenant complaints in stabilized buildings. Hazardous violations increased most in lower-income neighborhoods and neighborhoods with higher non-white population shares. I also found that landlords reduced apartment renovations in stabilized buildings while combining apartments, which let them reset the regulated rent of the combined unit until the state closed that route in November 2023. These findings support a long-standing prediction, going back to Friedman and Stigler (1946), that rent regulation erodes housing quality by weakening landlords' incentive to maintain their buildings.

A full welfare analysis of HSTPA would need to weigh the costs documented here against the benefits of stronger rent protections and reduced displacement risk for tenants, along with the broader effects of rent control on supply and misallocation. These results have immediate policy relevance. On June 25, 2026, the New York City Rent Guidelines Board adopted a rent freeze for one- and two-year leases in rent-stabilized apartments beginning in October 2026.[21] Rent regulation remains under active debate in many other cities. The findings suggest that policymakers should seriously weigh the potential harms of these policies and consider complementary measures, whether by increasing incentives for maintenance and investment or by increasing code enforcement. Quantifying the tenant-side benefits against the costs documented here is the natural next step toward assessing the policy as a whole.

[21]The adopted guidelines set a 0 percent adjustment for one- and two-year leases commencing between October 1, 2026, and September 30, 2027. See the New York City Rent Guidelines Board's *Adopted Summary of Guidelines (2026–27)*.

Figures and Tables

Table 1: Sample Construction

Sample Restriction	Buildings	Dropped
All NYC lots (PLUTO)	858,977	—
Multifamily (≥6 units)	64,621	794,356
Valid year built	63,659	962
Built 1990 or earlier	56,510	7,149
Excl. NYCHA & pre-HSTPA HPD agreements	54,525	1,985
Excl. pre-HSTPA operating subsidies (Furman SHD)	53,616	909
Excl. on 2024 DHCR list only	49,435	4,181
Excl. condos (R-class)	47,152	2,283
Excl. co-ops	41,256	5,896

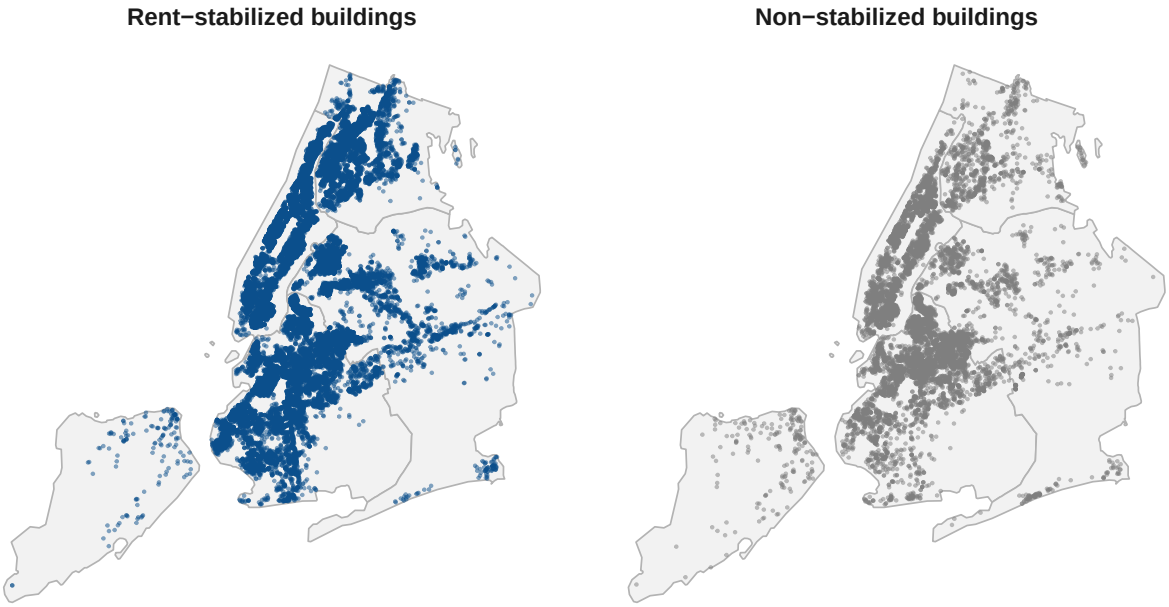
Note: Buildings remaining after each sample restriction, with the number dropped at each step. NYCHA is New York City Housing Authority public housing. HPD exclusions are production agreements closed before June 14, 2019. That source starts in 2014 and does not cover all older subsidized housing. Co-ops are PLUTO building classes C6, C8, CC, D0, D4, DC and H7.

Table 2: Summary Statistics

	Stabilized		Non-Stabilized	
	Mean	SD	Mean	SD
Panel A: Building Characteristics				
Total Units	26.7	41.5	17.0	52.8
Year Built	1921	18	1920	24
Panel B: Pre-Period Outcomes (monthly)				
Class C Violations	0.161	0.409	0.061	0.294
Class B Violations	0.466	1.146	0.155	0.785
HPD Complaints	0.563	2.026	0.166	0.762
N Buildings	31,066		10,190	

Note: Sample consists of 41,256 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, lots with a pre-HSTPA operating subsidy, condominiums, and co-ops. Rent stabilization status is determined from the 2017 DHCR registry. Violation and complaint data are from NYC HPD Open Data. Project-filing data are from NYC DOB Open Data. Outcome variables are measured at the building-month level during the pre-HSTPA period (July 2016 through June 2019). Class C violations are immediately hazardous conditions.

Figure 1: Geographic Distribution of Sample Buildings by Rent Stabilization Status



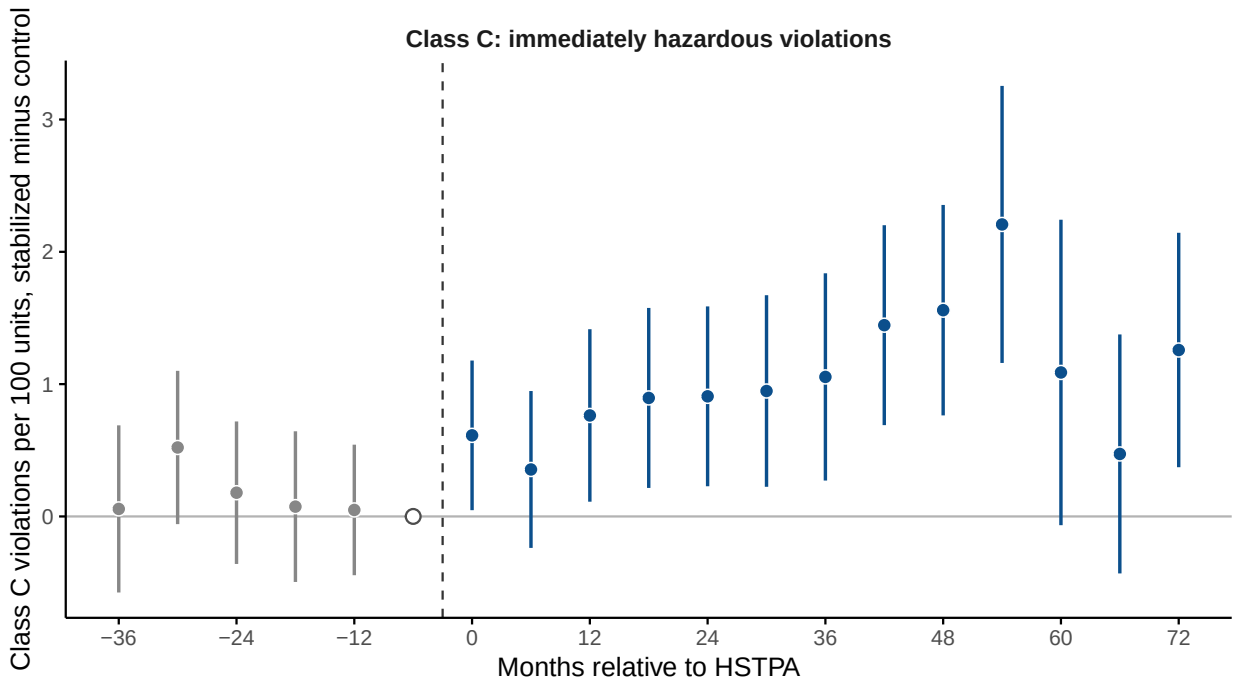
Note: Blue points are rent-stabilized buildings and gray points non-stabilized buildings, for every analysis-sample building with a valid PLUTO lot centroid. The analysis sample excludes NYCHA public housing, lots with a pre-HSTPA operating subsidy, condominiums, and co-ops, and contains 31,066 rent-stabilized and 10,190 non-stabilized buildings. Stabilization status comes from the 2017 DHCR registry.

Table 3: Effect of HSTPA on Building Outcomes

	Class A	Class B	Class C	Complaints
Stabilized × Post	-0.263**	0.335	0.897***	1.069***
	(0.128)	(0.341)	(0.208)	(0.361)
Baseline mean	4.29	10.45	3.61	12.62
Buildings	41,256	41,256	41,256	41,256
Observations	783,864	783,864	783,864	783,864

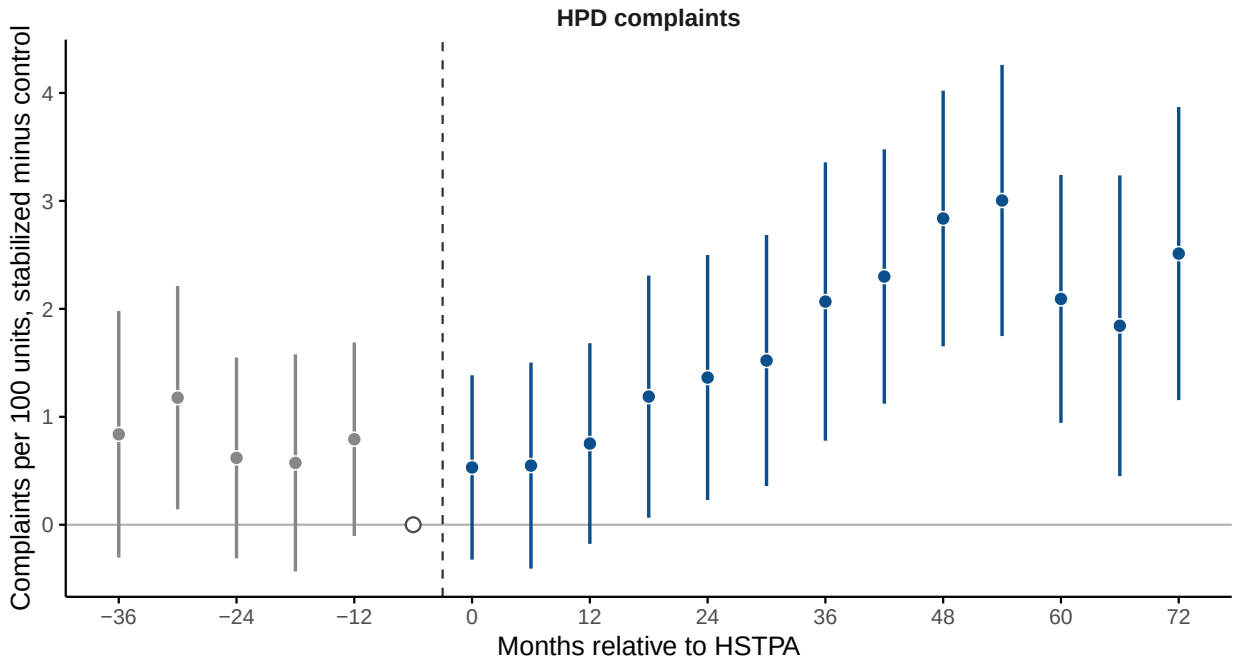
Note: This table tests whether HSTPA affected housing quality and investment outcomes in rent-stabilized buildings. Sample consists of 41,256 NYC buildings with six or more residential units built in 1990 or earlier, excluding NYCHA public housing, lots with a pre-HSTPA operating subsidy, condominiums, and co-ops. Data span July 2016 through December 2025 in 6-month periods. Reported coefficients are pooled difference-in-differences estimates on Stabilized × Post. Outcomes are per 100 units, and regressions are weighted by the number of residential units, so estimates describe the average unit rather than the average building. Specification includes building fixed effects plus tract × period, size-bin × period, decade × period, building-type × period, and baseline-quality-group × period fixed effects. "Baseline mean" is the pre-period (July 2016–June 2019) mean for stabilized buildings. Standard errors clustered at the building level in parentheses. Raw-count results giving each building equal weight appear in Table A1.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 2: Event Study: Effect of HSTPA on Immediately Hazardous (Class C) Violations



Note: Sample as in Table 3. Figure plots coefficients β_k on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019 (period $k = -1$). The outcome is Class C (immediately hazardous) HPD violations per 100 units. Appendix Figure A3 shows the same event study for Class B and Class A violations. Specification and weights as in Table 3. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Figure 3: Event Study: Effect of HSTPA on HPD Complaints



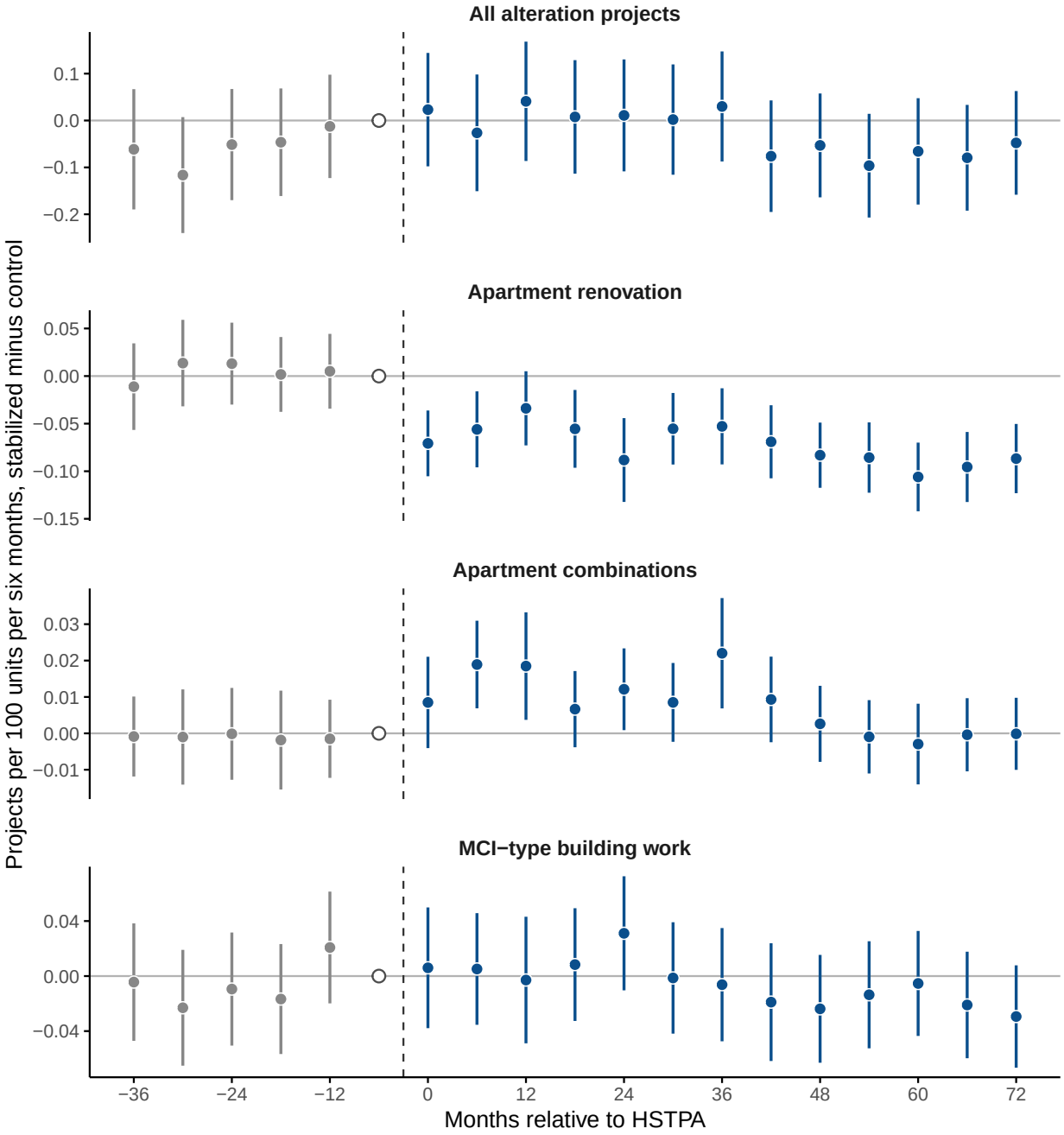
Note: Sample as in Table 3. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. Complaints are per 100 units. Specification and weights as in Table 3. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Table 4: Effect of HSTPA on Alteration Projects by Type of Work

	Apartment renovation	Apartment combinations	MCI-type work	Commercial (placebo)
Stabilized × Post	−0.0759***	0.0088***	−0.0001	−0.0036
	(0.0098)	(0.0022)	(0.0074)	(0.0055)
% of baseline	-46.3	159.4	-0.1	-8.2
Baseline mean	0.1641	0.0055	0.1189	0.0445
Pre-trend test (p)	0.90	1.00	0.33	0.28
Buildings	41,256	41,256	41,256	41,256

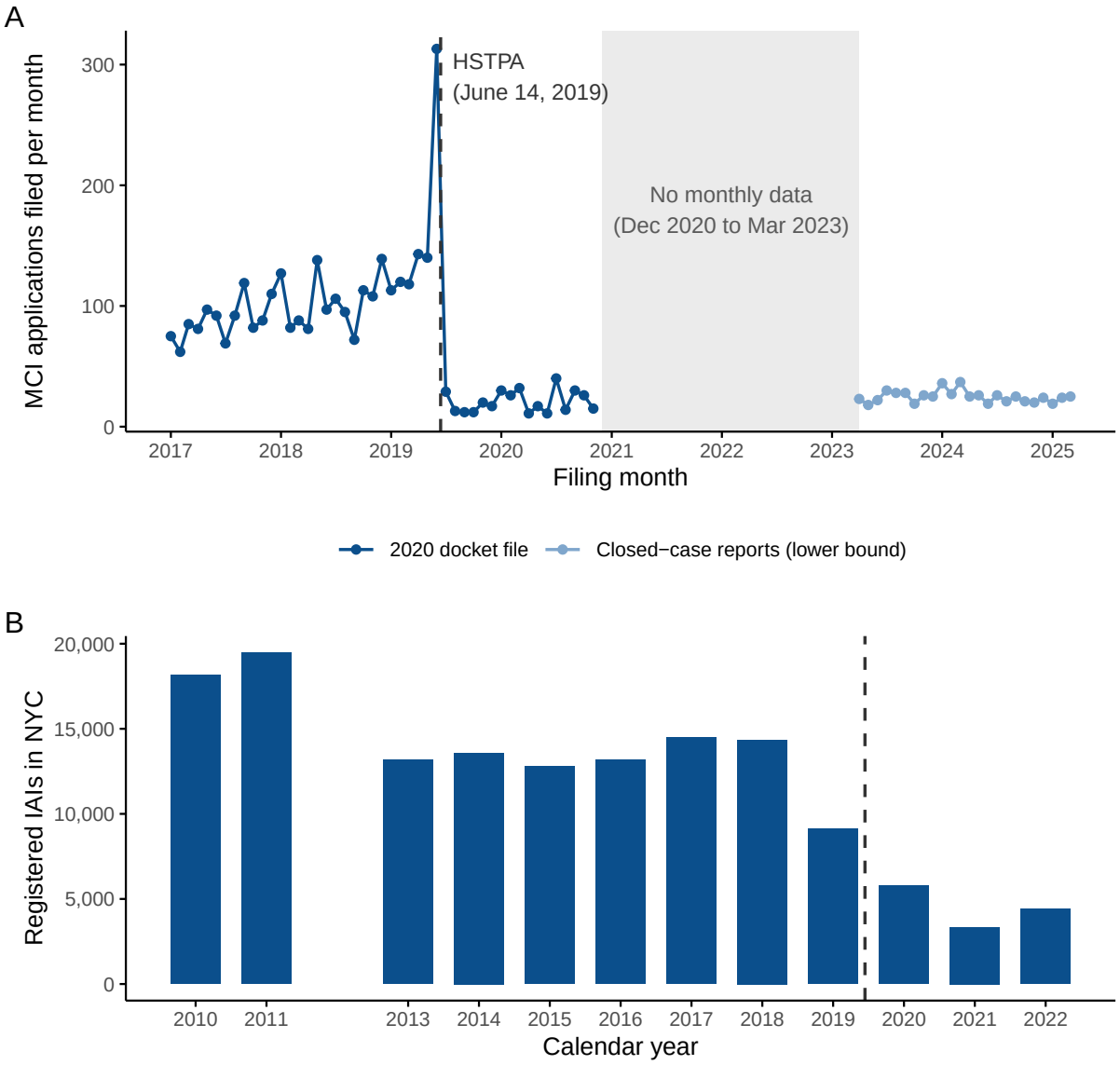
Note: Pooled difference-in-differences estimates of HSTPA's effect on DOB alteration projects per 100 residential units per six months. Columns are the type of work a language model (Claude Haiku 5.5) assigns to each filing's description. Apartment renovation is work in or for identified apartments. Apartment combinations are renovations that merge apartments or parts of apartments. MCI-type work is building-level work on items in DHCR's useful-life schedule for Major Capital Improvements (roofs, facades, windows, boilers, elevators and similar). Commercial space is work on stores, offices, and other non-residential space in the building, which HSTPA did not affect. Filings the model flags as companions to another job are dropped, so a project filed as several jobs counts once. Sample, specification and weights as in Table 3. "Baseline mean" is the July 2016–June 2019 mean for stabilized buildings. "Pre-trend test" is the p-value of a joint Wald test that the pre-HSTPA event-study coefficients are zero. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 4: Event Study: Effect of HSTPA on Alteration Projects by Type of Work



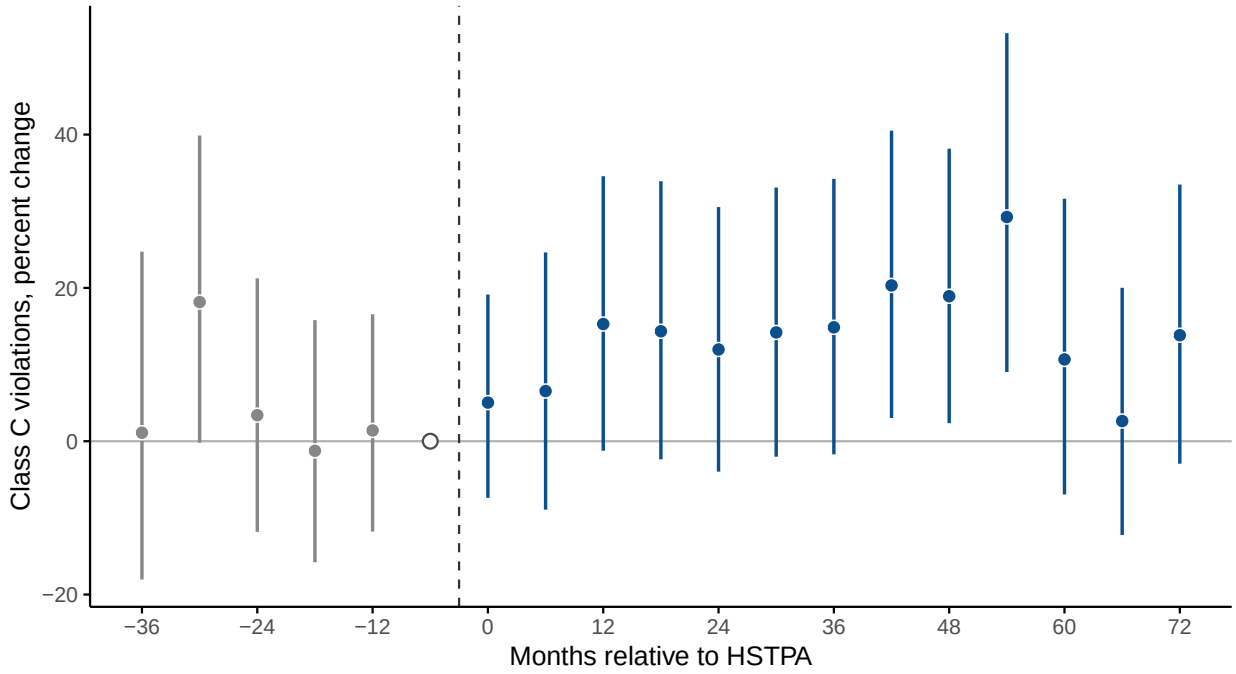
Note: The top panel is every alteration project without a unit-count change, per 100 residential units per six months. The three panels below are one type of project each. Types are assigned from the filing text by a language model (Claude Haiku 5.5) and defined in the note to Table 4. Filings the model flags as companions to another job are dropped from those three, so a project filed as several jobs counts once. Points are coefficients on interactions between a rent-stabilized indicator and 6-month period indicators, relative to January–June 2019 (open circle). Sample, specification and weights as in Table 3. The dashed line marks HSTPA passage (June 2019). Error bars show 95 percent confidence intervals, with standard errors clustered at the building level.

Figure 5: MCI Applications and Registered IAIs Over Time



Note: Panel A shows Major Capital Improvement (MCI) applications filed with DHCR per month in New York City. MCIs are building-wide improvements (boilers, roofs, elevators) for which landlords of rent-stabilized buildings can seek rent increases. The darker series, January 2017 through November 2020, counts applications by filing date in DHCR case records obtained by Winnie Shen through a FOIL request and distributed through NYCDB, excluding tax abatement cases. The lighter series, April 2023 through March 2025, counts applications by filing month in DHCR's monthly closed-case reports, which cover applications decided by August 2026. No monthly data exist for December 2020 through March 2023 (shaded). Panel B shows Individual Apartment Improvements (IAIs) registered with DHCR for New York City apartments by calendar year, 2010 through 2022, from the Rent Guidelines Board's 2023 explanatory statement (p. 37), which reports no figure for 2012. The 2019 bar mixes months before and after HSTPA. Dashed lines mark HSTPA passage (June 14, 2019).

Figure 6: PPML Event Study: Effect of HSTPA on Immediately Hazardous Violations



Note: Proportional effects from a PPML event study of raw six-month Class C violation counts, without unit weights. Points show $100[\exp(\widehat{\beta}_k) - 1]$ relative to January–June 2019 (open circle, normalized to zero). Sample and specification as in Table 3, except that PPML drops buildings whose outcome is always zero or otherwise perfectly predicted by the fixed effects. Periods are plotted at their starting month, and the dashed line marks HSTPA passage (June 2019). Error bars transform the endpoints of 95 percent pointwise confidence intervals, with standard errors clustered at the building level.

Table 5: Heterogeneity in Class C Violation Effects

Dimension	Group	Buildings	Baseline mean	Effect	Equal-effects p-value
Borough	Manhattan	13,722	2.69	0.605***	0.357
				(0.160)	
	Bronx	5,382	6.17	2.187**	
				(0.949)	
	Brooklyn	15,287	4.55	0.869**	
				(0.347)	
	Queens	6,534	1.20	0.681	
				(0.475)	
Nbhd Income	Low	12,796	5.53	1.216**	0.013
				(0.485)	
	Middle	10,410	3.25	0.935**	
				(0.398)	
	High	12,693	1.41	0.098	
				(0.144)	
Non-white	Low	12,021	1.28	0.281*	0.113
				(0.161)	
	Middle	11,964	3.05	0.336	
				(0.273)	
	High	11,929	5.78	1.387***	
				(0.524)	

Note: Group-specific effects of HSTPA on Class C violations, with each dimension's groups estimated jointly in one pooled difference-in-differences regression. "Effect" is the post-HSTPA change in stabilized buildings relative to non-stabilized buildings within the group. "Buildings" is the number of buildings in the group and "Baseline mean" the unit-weighted pre-HSTPA mean among its stabilized buildings. The equal-effects p-value is a Wald test that the effects within a dimension are equal. Neighborhood income and non-white population share are tract-level 2015–2019 ACS five-year estimates, split at terciles. Sample, specification and weights as in Table 3, restricted to buildings with a group classification. The borough comparison excludes Staten Island. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: HSTPA Effects by Pre-Policy Mortgage Exposure

Mortgage-exposure group	HSTPA effect	Difference from low-exposure group
No qualifying ACRIS mortgage	0.606**	0.252
	(0.266)	(0.390)
Low mortgage exposure	0.355	—
	(0.321)	
Middle mortgage exposure	1.319***	0.965*
	(0.458)	(0.528)
High mortgage exposure	1.537***	1.182**
	(0.511)	(0.577)
Equal effects across groups (p-value)	0.104	

Note: Group-specific effects of HSTPA on Class C violations by pre-policy mortgage exposure, estimated jointly in one pooled difference-in-differences regression. Mortgage exposure is the amount on the building's most recent ACRIS mortgage recorded between January 2010 and June 13, 2019, divided by its market value on the FY2020 tax roll and winsorized at the 99th percentile. Low, middle, and high are terciles of this ratio among buildings with a mortgage on record. Buildings with no mortgage on record form a fourth group, and buildings with a mortgage document but no usable amount are excluded. "HSTPA effect" is the post-HSTPA change in stabilized buildings relative to non-stabilized buildings in the same group. "Difference from low-exposure group" compares each estimate with the low-exposure tercile. The equal-effects p-value is a Wald test that the four effects are equal. Sample, specification and weights as in Table 3, excluding Staten Island, whose property records are not in ACRIS, and adding mortgage-exposure-group $\times$ period fixed effects. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

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A Appendix Figures and Tables

Table A1: Effect of HSTPA on Building-Weighted Count Outcomes

	Class A	Class B	Class C	Complaints
Stabilized × Post	-0.004	0.090*	0.118***	0.194***
	(0.019)	(0.052)	(0.033)	(0.060)
Baseline mean	1.15	2.80	0.97	3.38
Buildings	41,256	41,256	41,256	41,256
Observations	783,864	783,864	783,864	783,864

Note: This table reports pooled difference-in-differences estimates using raw 6-month counts per building. The coefficient is the change in the number of violations, complaints, or alteration-project initiations for the average stabilized building after HSTPA, relative to comparable non-stabilized buildings. Sample as in Table 3. Each building is weighted equally, in contrast to the main specification, where outcomes are per 100 units and regressions are weighted by the number of residential units. Specification as in Table 3. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A2: Robustness to Alternative Fixed Effects Specifications

	Bldg+Time	+Tract	+Size	+Decade	+Class	+Baseline
Stabilized × Post	1.923***	1.160***	1.106***	1.089***	0.982***	0.897***
	(0.175)	(0.186)	(0.190)	(0.197)	(0.205)	(0.208)
Calendar period	Yes	Yes	Yes	Yes	Yes	Yes
Building	Yes	Yes	Yes	Yes	Yes	Yes
Tract × Time		Yes	Yes	Yes	Yes	Yes
Size Bin × Time			Yes	Yes	Yes	Yes
Decade Built × Time				Yes	Yes	Yes
Building Class × Time					Yes	Yes
Baseline Outcome × Time						Yes
Buildings	41,256	41,256	41,256	41,256	41,256	41,256
Pre-trend joint p-value	<0.001	0.001	0.004	0.003	0.013	0.253
Observations	783,864	783,864	783,864	783,864	783,864	783,864

Note: Pooled post-period coefficients from difference-in-differences regressions of Class C violations per 100 units on Stabilized × Post, where Post is one for periods after June 2019. Sample and weights as in Table 3. Columns add fixed effects one set at a time, from building plus calendar-period effects (column 1) to the full specification of Table 3 (column 6). "Pre-trend joint p-value" is from a Wald test that the pre-period event-study coefficients are jointly zero. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A3: Level and Poisson Estimates for Count Outcomes

Panel A: Overall count outcomes

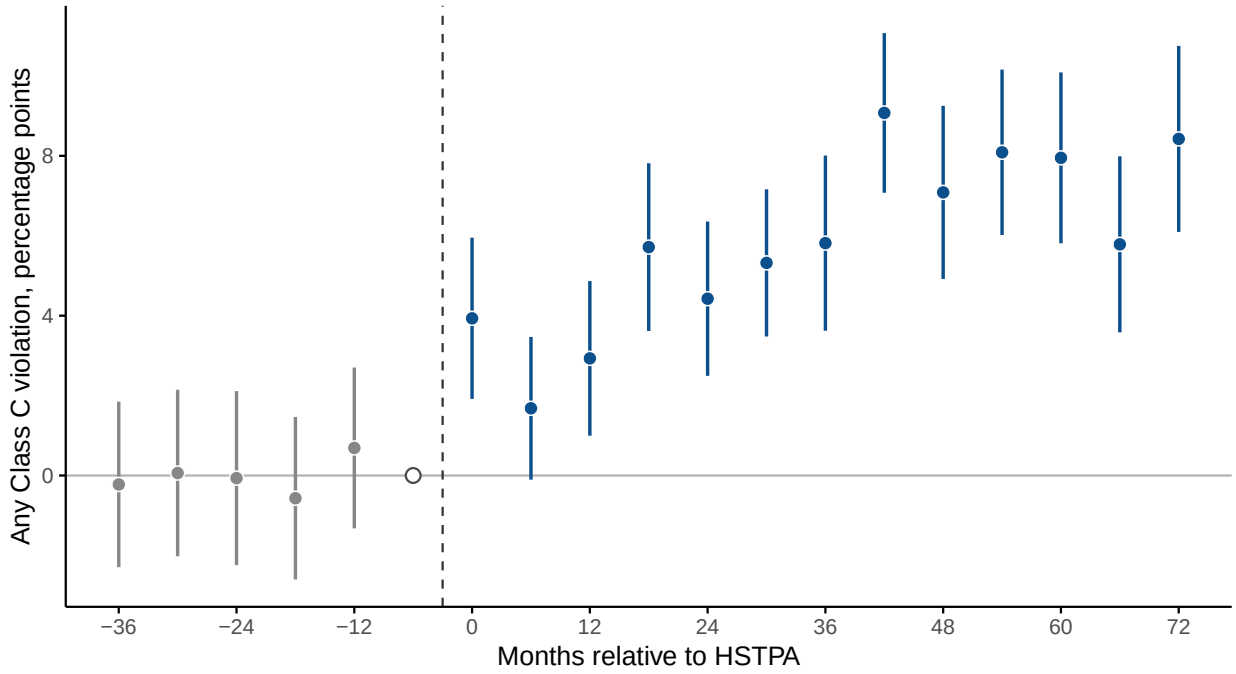
	Class C	Class B	Complaints
Main level estimate	0.897***	0.335	1.069***
	(0.208)	(0.341)	(0.361)
Percent of baseline	24.8%	3.2%	8.5%
Level estimate, PPML sample	0.530	-0.010	0.691
	(0.342)	(0.566)	(0.447)
Percent of baseline	12.9%	-0.1%	5.2%
PPML coefficient	0.096**	0.023	0.037
	(0.048)	(0.043)	(0.035)
PPML implied effect	10.1%	2.3%	3.8%
PPML pre-trend joint p-value	0.260	0.409	0.031
Buildings, PPML sample	27,692	27,756	34,175
Buildings, full sample	41,256	41,256	41,256

Panel B: Class C effects by baseline condition

	Baseline zero	Baseline low+	Baseline high
Baseline mean	0.00	1.25	8.88
Level estimate	1.147***	1.398***	0.194
	(0.107)	(0.412)	(1.292)
PPML coefficient	—	0.263***	0.079
	—	(0.050)	(0.062)
PPML implied effect	—	30.1%	8.2%
Joint pre-trend p-value	—	0.223	0.255
Buildings	23,565	9,513	8,178
Equality of low-positive and high PPML coefficients: $p = 0.021$			

Note: Panel A compares the main level specification, the same specification re-estimated on the buildings PPML retains, and PPML. Level specifications are per 100 units with unit weights, as in Table 3. PPML uses raw six-month counts without weights and the same fixed effects. The PPML implied effect is $100[\exp(\widehat{\beta}) - 1]$. "Pre-trend joint p-value" is from a Wald test that the pre-HSTPA event-study coefficients are jointly zero. Panel B splits buildings into those with no baseline Class C violations, those at or below the median among buildings with positive baseline violations, and those above it. Level specifications are estimated within each group and omit baseline-group × period fixed effects. The last row tests whether the PPML coefficients of the two positive-baseline groups are equal. Standard errors clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure A1: Event Study: Buildings Recording Any Immediately Hazardous Violation



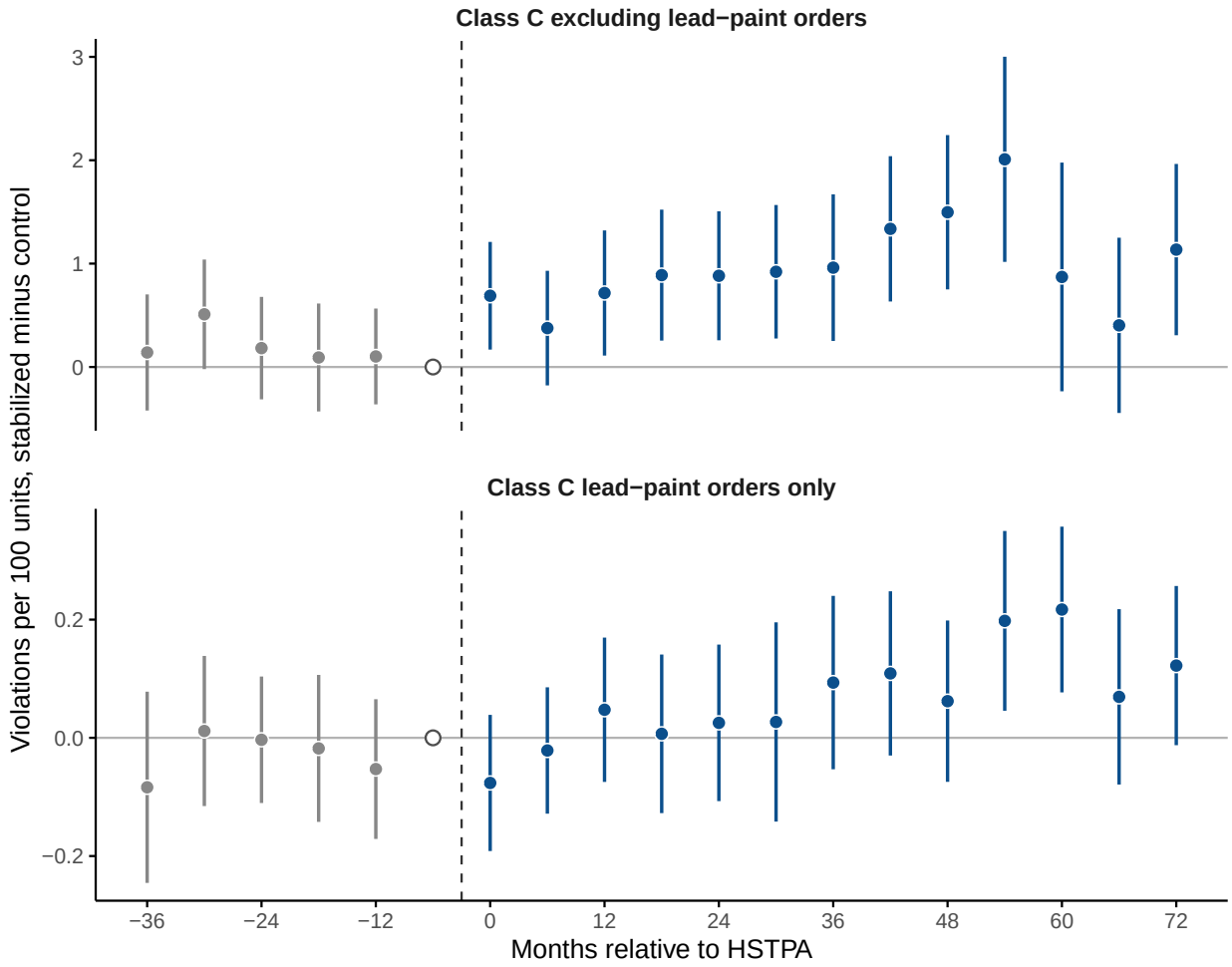
Note: The outcome is an indicator for at least one recorded Class C violation in a six-month period, so coefficients are percentage points. Regressions are weighted by baseline residential units, so the outcome is the share of baseline apartments in buildings recording at least one violation, not the share of apartments with a violation. Sample and specification as in Table 3. The open point is January–June 2019, normalized to zero. Periods are plotted at their starting month, and the dashed line marks HSTPA passage (June 2019). Error bars show 95 percent pointwise confidence intervals, with standard errors clustered at the building level.

Table A4: COVID-19 Period Robustness

Outcome	Main sample	Excl. 2020–21
Class C Violations	0.897***	1.036***
	(0.208)	(0.243)
All Complaints	1.069***	1.412***
	(0.361)	(0.389)
Buildings	41,256	41,256

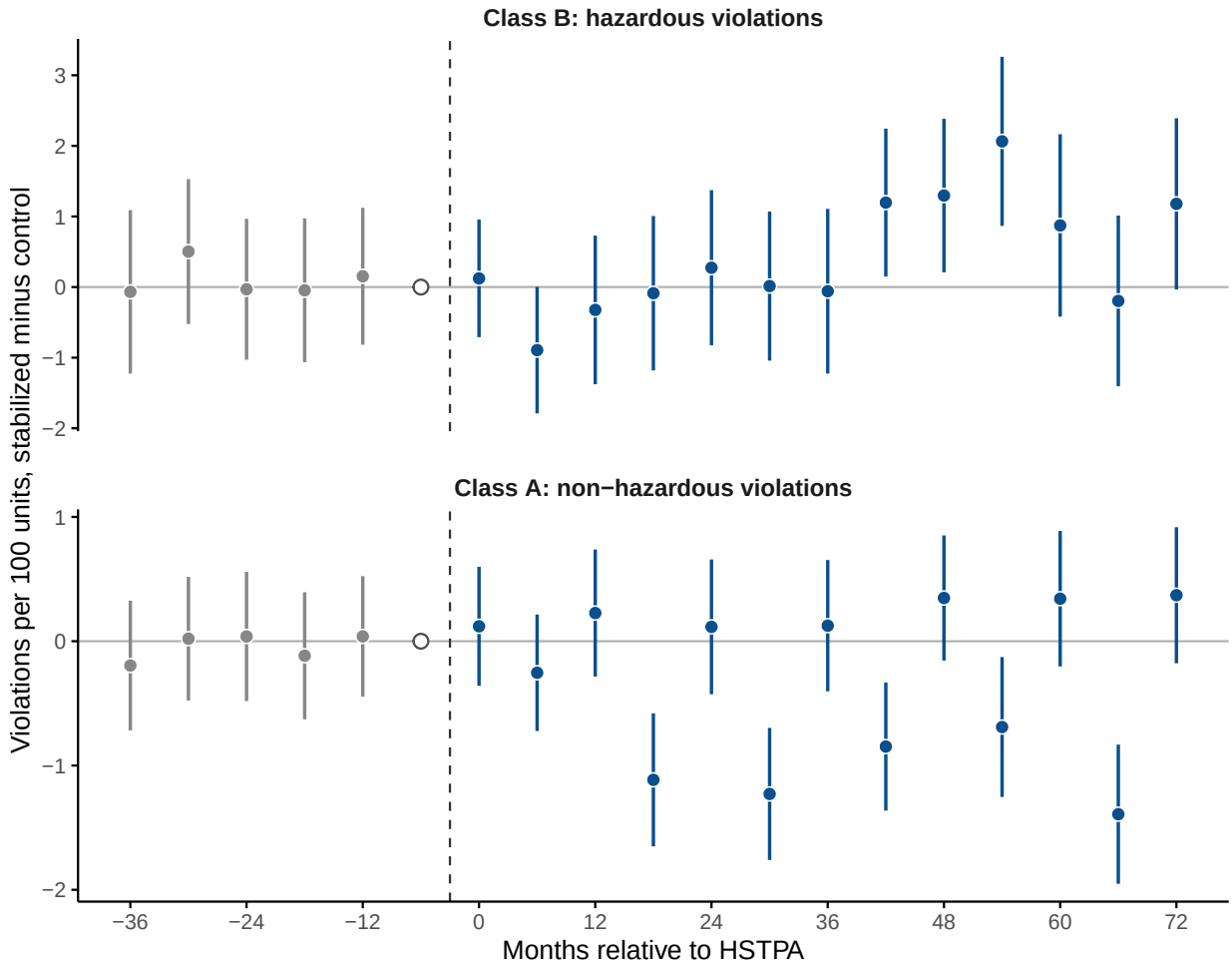
Note: Comparison of main results including versus excluding the COVID-19 pandemic period (January 2020–December 2021). Sample as in Table 3. Specification as in Table 3. All outcomes are per 100 units, with regressions weighted by the number of residential units. Standard errors clustered at building level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure A2: Event Study: Class C Violations Excluding and Including Only Lead-Paint Orders



Note: Each panel plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019 (open circle). The top panel's outcome is Class C violations per 100 units excluding lead-paint orders. The bottom panel's outcome is lead-paint orders alone, HPD violations issued under Local Law 1 of 2004 (order numbers 614 and 616 through 626). The two outcomes sum to the Class C series in Figure 2. The vertical scales differ. Sample as in Table 3. Specification and weights as in Table 3. The vertical dashed line marks HSTPA passage (June 2019). Error bars show 95 percent confidence intervals with standard errors clustered at the building level.

Figure A3: Event Study: Effect of HSTPA on Class B and Class A Violations



Note: Each panel plots coefficients on interactions between a rent-stabilized indicator and 6-month period indicators, relative to January–June 2019 (open circle), for HPD violations per 100 units in one hazard class. Class B is hazardous, with 30 days to correct. Class A is non-hazardous, with 90 days. Figure 2 is the same event study for Class C, and Table A6 reports the pooled estimates for all three classes. Sample, specification and weights as in Table 3. The dashed line marks HSTPA passage (June 2019). Error bars show 95 percent confidence intervals, with standard errors clustered at the building level.

Table A5: HSTPA Effects on Class C Violations: Lead-Paint Orders and All Other

	Total	Ex-lead	Lead-only
Stabilized × Post	0.897***	0.805***	0.092***
	(0.208)	(0.193)	(0.033)
% of baseline	24.8%	26.0%	17.6%
Baseline mean (stabilized)	3.613	3.091	0.522
Pre-trend joint p-value	0.253	0.266	0.748
Buildings	41,256	41,256	41,256

Note: This table splits Class C violations into lead-paint orders and all other Class C violations and estimates the main specification on each part. Lead-paint orders are HPD violations issued under Local Law 1 of 2004 (order numbers 614 and 616 through 626). They cover lead hazards and lead recordkeeping and are Class C by statute. Sample as in Table 3. Each column is a separate pooled difference-in-differences regression with the specification and weights of Table 3. The two component estimates sum to the total up to rounding. "Baseline mean" is the pre-period (July 2016–June 2019) mean for stabilized buildings, and "Pre-trend joint p-value" is from a Wald test that the pre-HSTPA event-study coefficients are jointly zero. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A6: HSTPA Effects by Violation Severity Class

Class	Severity	Deadline	Effect	% of baseline
C	Immediately hazardous	Varies	0.897***	25%
			(0.208)	
B	Hazardous	30 days	0.335	3%
			(0.341)	
A	Non-hazardous	90 days	-0.263**	-6%
			(0.128)	

Note: HSTPA effects by violation severity class, from pooled difference-in-differences regressions of violations per 100 units with the sample, specification and weights of Table 3. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B Classifying Alteration Filings

To study landlords' investment responses to HSTPA at a more granular level, I use a large language model (LLM), Claude Haiku 5.5, to classify the free-text description on each Department of Buildings alteration filing. I classify each filing into the ten categories listed in Section 3. For each filing, the model receives the job description and, when available, the floor number and apartment unit fields. Filings with identical text, floor, and apartment fields are classified only once. Ultimately, I classify 111,279 distinct text inputs.

Classification rules. The model assigns each filing to one of ten categories:

1. *Apartment renovation* is work inside or for an identified apartment.

2. *Interior work with no stated location* includes partitions, finishes, kitchens or baths where nothing says whether it was in an apartment, a store, or a shared space.

3. *Building systems* include heating, hot water, plumbing, gas, electrical, elevator, or fire-safety work serving the building.

4. *Envelope* is the facade, roof, windows, parapets, or fire escapes.

5. *Structural* is beams, slabs, or steel.

6. *Common areas* are lobbies, halls, stairs, cellars, and laundry or amenity rooms.

7. *Site work* is fences, courtyards, walkways, driveways, and sidewalks.

8. *Commercial space* is a store, office, or other non-residential space.

9. *Not investment* covers sidewalk sheds, scaffolds, inspections, legalizations of past work, and filings with no physical work.

10. *Unclear* is empty or generic text such as "general construction."

The model also returns a tag for each work type the filing names, from a fixed list of 23 that includes items on DHCR's Useful Life Schedule (see Section 3), such as windows, roofs, facades, boilers, elevators, and apartment combinations. It also flags filings that are a companion to another application, such as when a plumbing job is "filed in conjunction with" a renovation. Finally, it flags filings prompted by something other than the owner's own choice, such as a violation, an order, fire damage, or a facade-inspection cycle. Table A7 lists eleven filings from the sample with the category and flags the model assigned.

Validation. I validate the model using more sophisticated large language models to review the results of my classification. I draw a random sample of 265 filings, stratified by the building's rent-stabilization status and whether the filing is before or after HSTPA. I oversample groups I found difficult to classify, companion filings and interior work with no stated location, and then weight the sample to represent all projects. Then, I have two separate instances of Claude Fable 5.1, a more sophisticated and resource-intensive large language model, classify each filing independently using the classification rules. When the models disagree, a third Claude Fable 5.1 instance adjudicates.

Table A8 compares the classifier with the Fable labels for each category. Precision is the share of filings the classifier places in a category that the Fable labels also place there, while recall is the share of filings that the Fable labels place in a category that the classifier also places there.

Table A7: Eleven Filings and How the Classifier Read Them

Filing description (verbatim)	Floor field	Category	Note
Interior renovation as shown on plans filed herewith.	Floor 3	Apartment renovation	one numbered floor places the work in an apartment
Interior renovation as shown on plans filed herewith.	Floors 1 through 5	Interior work, no location	several floors named, so no apartment is identified
Interior renovation as shown on plans filed herewith.	Cellar	Common area	cellar is shared space
INTERIOR RENOVATION AS PER PLANS FILED HEREWITH. GEN. CONSTR.	—	Unclear	no floor field (BIS filing), generic text
COMBINATION OF APARTMENTS 12G AND 12H AS PER TPPN #3/97. GEN. CONSTR.	—	Apartment renovation	combination tag
REPLACE BOILER AND GAS/OIL BURNER.	Cellar	Building system	MCI-eligible item
FACADE REPAIRS AS PER PLANS FILED HEREWITH. NO CHANGE IN USE, EGRESS OR OCCUPANCY. FACADE	—	Envelope	MCI-eligible item
INSTALLATION OF KITCHEN FIRE SUPPRESSION SYSTEM. NO CHANGE IN USE,EGRESS OR OCCUPANCY.	—	Commercial space	
INSTALLATION OF TEMPORARY SIDEWALK SHED AS PER PLANS. NO CHANGE IN USE, EGRESS OR OCCUPANCY.	Open Space	Not investment	
FILING HEREWITH FOR INSTALLATION OF FENCE. FILED IN CONJUNCTION WITH WALL REMOVAL LLW# 092285.	—	Site work	companion flag: rides on another filing
FILE SUSPENDED SCAFFOLD OUTRIGGER SYSTEMS FOR FACADE RESTORATION AND INSPECTION.	—	Not investment	forced flag: facade-inspection cycle

Note: Each row shows the most frequent description among sample filings of one kind, such as envelope work or companion filings. The first three rows share a description to show how the floor field changes the category. "Floor field" is a field in the DOB NOW data only, and is missing from the BIS filings data. The combination tag, the companion flag, and the forced flag are classifications produced by the model.

Table A8: Classifier Validation

	Share of projects	Precision	Recall
Apartment renovation	35.2%	99.6%	98.2%
plus interior work with no location	38.2%	98.3%	97.2%
Building systems or envelope	16.5%	91.2%	99.0%
Commercial space	6.5%	99.6%	96.1%
Apartment combinations†	2.5%	93.8%	100.0%

Note: This table scores the classifier against the Fable labels on 265 filings, weighted to all 152,751 projects. "Share of projects" is the Fable labels' weighted share in each category. †Scored on a separate sample of 140 projects drawn to include apartment combinations, which are too rare in the test set to score.

Constructing the outcomes. Using the LLM classifications and flags, I group the alteration projects into the four outcomes listed in Section 3, each counted per building and six-month period by the date of the initial filing:

1. *Apartment renovation* is the apartment-renovation category, excluding apartment combinations.

2. *Apartment combinations* are the apartment renovations that include the apartment combination tag.

3. *MCI-type work* is alterations on building systems, envelope, structural, common area, and site work categories and also carries a tag for an item on DHCR's Useful Life Schedule.

4. *Commercial space* is the commercial category and serves as a placebo.

The main estimates exclude filings flagged as companions to another application, so one job isn't counted twice.

C Landlord Matching and Portfolio Classification

This appendix describes how I identify buildings with a shared owner. The data are the HPD Registration Contacts and Multiple Dwelling Registrations files from NYC Open Data, downloaded in February 2026. I join registrations to tax lots by BBL. For each registration, I select one person as the owner: the individual owner when listed, or the joint owner, head officer, or officer otherwise. Contacts with placeholder names such as "#" are excluded. I standardize each name as first name, middle initial, and last name in uppercase, and I assign buildings linked to the same person to one owner group.

Name variants. Registrations sometimes omit a middle name or record only an initial. Exact matching can therefore assign the same person to separate owner groups. Matching

only on first and last names would recover some of these links but could also combine different people. I therefore merge two name variants only when both are linked to the same building and share a first and last name. I extend the same treatment to pairs of names linked to the same building whose character-level similarity is at least 85 percent (e.g., "Hirschfield" and "Hirscfield"). Names never linked to a common building remain separate owners, so the linkage may understate how many buildings some landlords hold.

Portfolios. I classify each owner's portfolio by the stabilization status of the analysis-sample buildings linked to that owner: all-stabilized if every linked building is rent-stabilized, all-market if none is, and mixed if the owner holds both. Of the 24,668 owners identified, 1,335 (5.4%) hold mixed portfolios, covering 11,665 buildings in the sample. Table A9 reports the counts by owner type. For the portfolio-spillover analysis in Section 5, I run baseline regressions after dropping buildings linked to owners with both stabilized and non-stabilized properties, as well as buildings I could not link to an owner.

Table A9: Landlord Portfolio Classification

Portfolio Type	Owners	Buildings
All-stabilized	14,399	22,275
All-market	8,934	6,106
Mixed	1,335	11,665
Total with owner info	24,668	40,046
Unknown (no owner match)	—	1,210

Note: Owners are identified from HPD registration contacts as described in this appendix. A portfolio is "all-stabilized" if all of an owner's linked sample buildings are rent-stabilized, "all-market" if none are, and "mixed" if the owner holds both. The Buildings column counts unique sample buildings in each portfolio type, not owner-building links. Buildings that could not be linked to an owner are excluded from the portfolio-spillover check in Section 5.